BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 285 · Network security, proxies and firewalls

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Can network packets travelling between the same two endpoints take different routes?

Q2. What is the transport-layer role of TCP?

Q3. What does a rule ‘allow if staff AND authenticated’ mean if authenticated is false?

Q4. Why can an open port be useful but also increase risk?

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I Do · Read new knowledge

Key vocabulary

firewall — Security control that permits or blocks network traffic according to rules.

forward proxy — Intermediary that makes requests on a client's behalf.

filter rule — A condition and action defining whether traffic is permitted.

cache — Stored copy of suitable data for reuse.

defence in depth — Use multiple complementary security controls rather than trusting one.

New learning

Check a firewall rule — A firewall applies configured rules to traffic according to properties such as direction, source/destination, ports and state. A basic rule example can allow TCP port 443 but deny 22.

Different request, different decision — A TCP connection to port 22 is denied by the illustrated policy. A firewall rule is not automatically a guarantee that the permitted service is safe.

A proxy is an intermediary — A forward proxy makes requests on behalf of clients, potentially enforcing policy or caching suitable resources. HTTPS content remains encrypted unless a controlled interception mechanism is in place.

Worked example

For an inbound rule set allowing TCP 443 and denying TCP 22, decide whether two connection requests are permitted.

1. 443 — An inbound TCP connection destined for port 443 matches the allow rule and is permitted.

2. 22 — An inbound TCP connection destined for port 22 matches the deny rule and is blocked.

3. Scope — An allow rule permits traffic; it does not prove that an application or connection is harmless.

We Do · Practise with guidance

A school uses a forward proxy to restrict browsing. Explain the route for a permitted request and for a blocked site.

Allowed resource

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The student's request reaches the proxy, which permits it and contacts the website; a suitable cached response may be used instead.

Blocked resource

If the proxy's policy denies the destination, it does not forward the request to that site; the student receives a blocking response.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

A firewall allows inbound TCP 443 but denies inbound TCP 22. Is an inbound connection to destination port 22 allowed? Explain. [2]

Question 2

Name and explain two different functions of a forward proxy in a school network. [4]

Question 3

Why is the claim ‘a firewall makes all allowed traffic safe’ incorrect? Explain two risks that can remain on allowed connections and identify two additional safeguards. [6]

Exit Ticket · Answer before checking

A forward proxy and firewall are identical because both can restrict access.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Yes, routing decisions and changing network conditions can lead to different paths.

Q2. A reliable, ordered byte stream between endpoints.

Q3. Deny; AND requires both conditions true.

Q4. It permits needed service connections, but exposes that service to network-originated attacks or probing.

You Do pages · Independent questions

Q1. [2] No, the connection is blocked [1], because inbound TCP destination port 22 matches the deny rule [1].

Q2. [4] Accept any two appropriately explained functions. Examples: apply browsing rules [1] to deny disallowed destinations before contacting them [1]; cache suitable responses [1] to reduce repeat external downloads [1]; or log request metadata [1] to support authorised monitoring or troubleshooting [1]. Maximum two functions × 2 marks.

Q3. [6] Allowed applications can contain exploitable vulnerabilities [1], with a relevant explanation such as an unpatched service being attacked on an allowed port [1]. Permitted traffic can also contain phishing or malicious content [1], which may still reach a browser or user [1]. Identify two additional safeguards with purpose, e.g. patching to close known flaws [1] and strong authentication or monitoring to reduce misuse or detect incidents [1]. Credit other valid distinct risks and defences.

Exit Ticket · Model answer

False. A proxy acts as an intermediary for requests; a firewall enforces rules on traffic.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed: